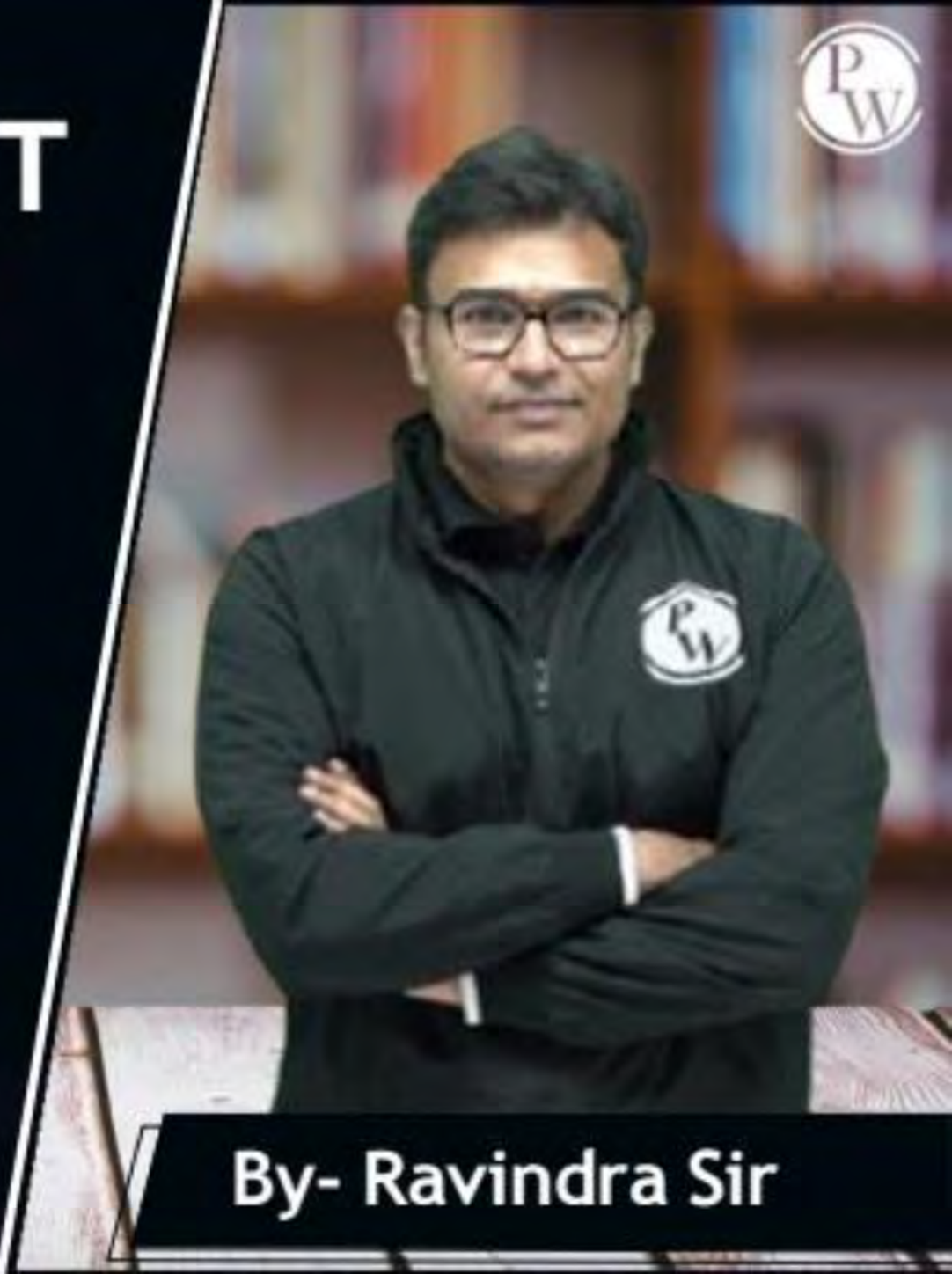
Computer Science & IT

COMPUTER NETWORK (CN)

Application Layer Protocols

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

FTP

Topics to be Covered



- Topic
- Topic
- Topic

HTTP

Security for FTP:-

The FTP was designed when Security was not a big issue. Although FTP required a password, it is sent in plain text (not encrypted), which it can be intercepted and used by any attacker. The data transfer also happens in plain text. To be secure, one can add a Secure Socket layer b/w the FTP application layer and TCP layer. In this case FTP is called SSL-FTP

WWW and HTTP:- (Hyper text transfer protocol)

The idea of web was first proposed by Tim Berners Lee in 1989. The commercial web started in 1900's.

The web today is a repository of information in which the documents are called web pages are distributed all over the world. web pages are linked together if they are related. The linking of webpages was achieved through using a concept called hyper text.

Today hypertext has been renamed as hypermedia.

To show that a webpage can be a text document
or an image or audio file or a video file.

Architecture of www :-

The www today is a distributed client-server service in which client using a browser can access a service using a server.

The service provided is distributed over many locations called sites. Each site holds one or more documents referred as webpages.

Each webpage can contain links to other webpages

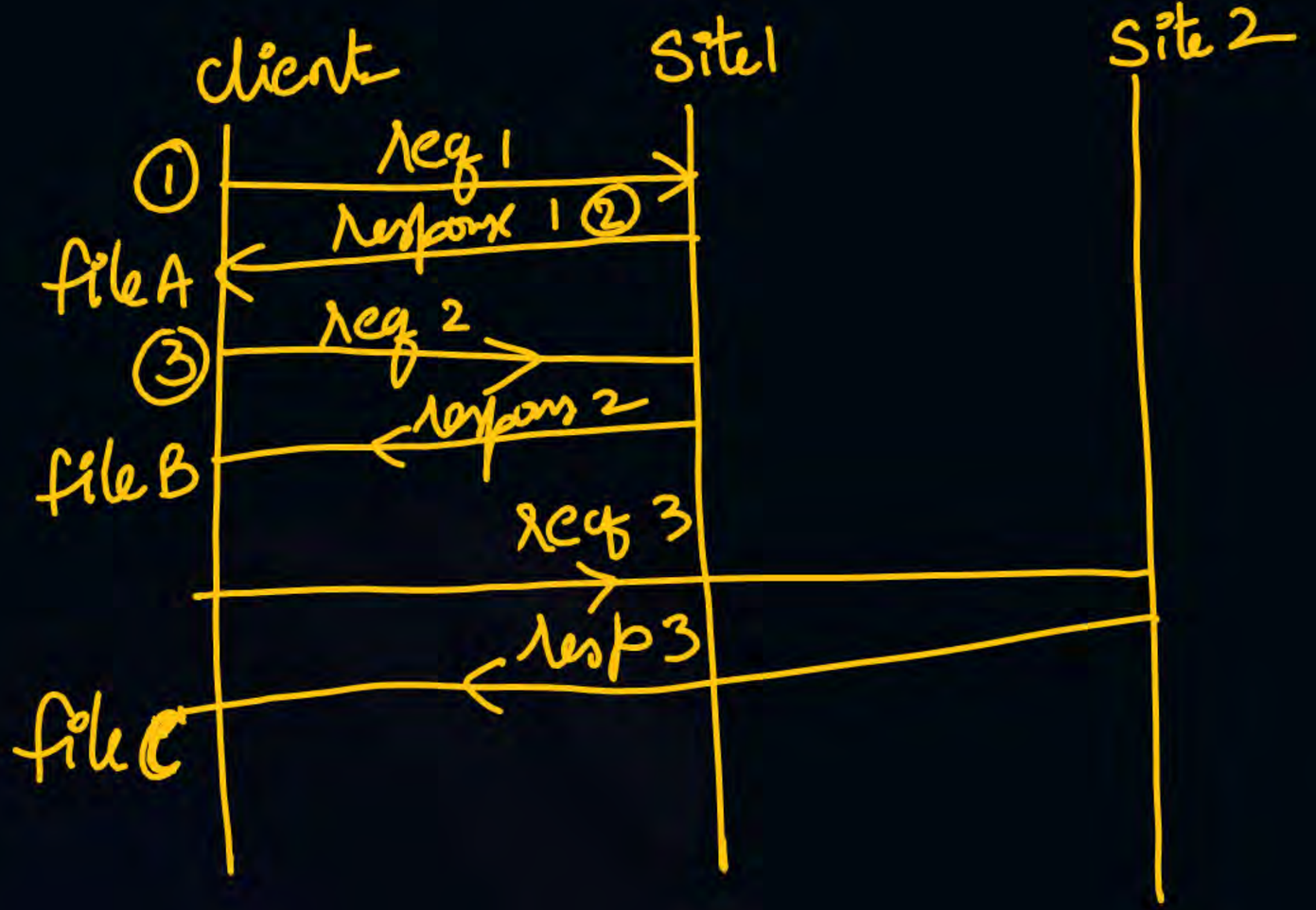
Webpages are of two types :-

a) Simple → has no links to other webpages

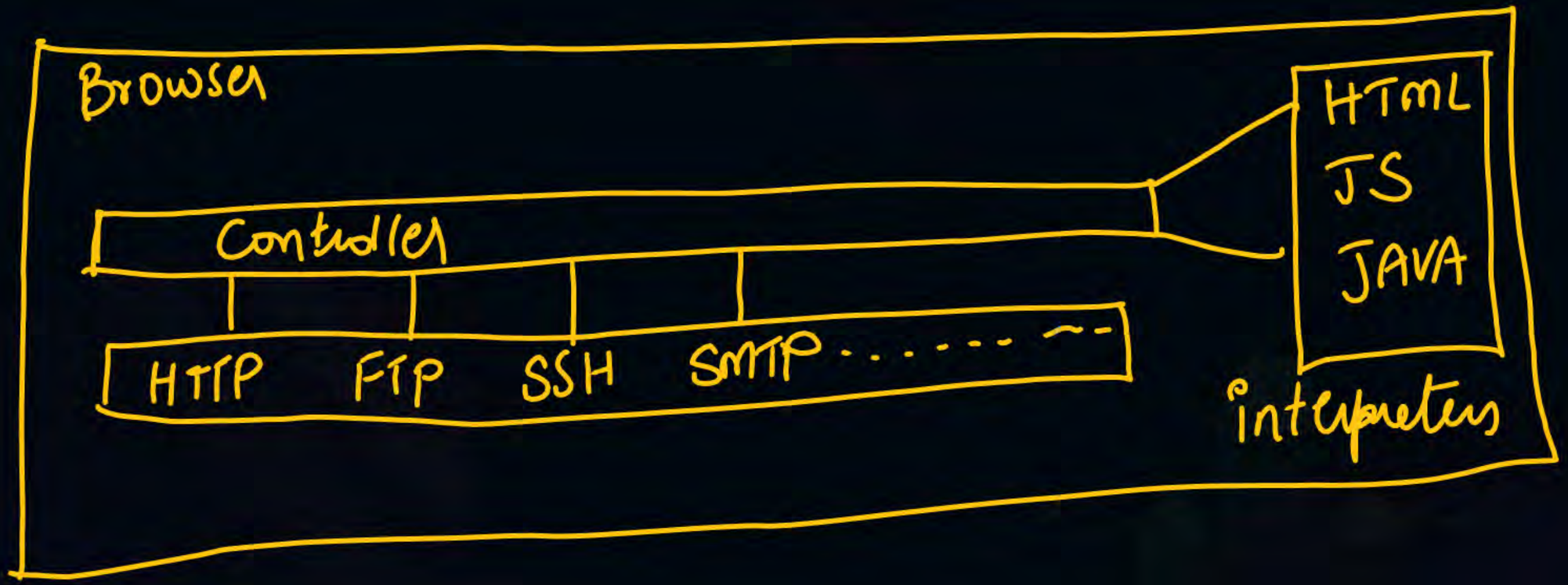
b) Composite → has one or more links to other webpages

Assume we need to retrieve a scientific document that contains one reference to another text file and one reference to a large image. The main document and the image are stored in two seperate files in the same site (fileA and File B), the referred text file is stored in another site (file C)




client
Site1
Site2
req 1
response 1
file A
req 2
response 2
file B
req 3
resp 3
file C

Web client (Browser) :-



The Controller receives i/p from the keyboard of the mouse and uses the client programs to access the data.
After a document has been accessed, the Controller uses one of the interpreter to to display the document on the screen
The client protocol can be one of the protocols such as HTTP, FTP, SSH, SMTP.--- The interpreter can be HTML, Java of java script, depending on the type of the document
Some commercial browsers are Internet explorer, Netscape navigator firefox of google chrome

Web Server:-

The webpages are stored at a server. Each time, a request arrives, the corresponding document is sent to the client. To improve efficiency, servers normally store requested file in a cache of memory. A server can also become more efficient through multithreading or multiprocessing

Uniform resource locator (URL) :-

A web page, as a file need to have a unique identifier to distinguish it from other webpages

To define a web page, we need identifiers

1. Host
2. PORT
3. PATH

Before defining the webpage, we need to tell browser what client server application we want to use, which is called protocol

Protocol :- The first identifier is abbreviation of the client-server program that we need in order to access the web page. Although most of the times the protocol is HTTP, we can also use other protocols such as FTP, SMTP. - - -

Host :- The host identifier can be the IP address of the server or the unique name given to the server

Ex :- 200.1.2.3
or
RBR.com

Port :- It is a 16 bit integer and is normally predefined for client server application
For example HTTP protocol has port 80.

Path:- The path identifies the location and the name of the file in underlying OS. In unix, the path is a set of directory names followed by the file name, all seperated by a slash.

Ex:- /top/next/last/myfile

URL format :-

Protocol://host/path → used most of the time

&

Protocol://host:port/path → when port no is needed

Web documents :-

The documents in www can be grouped into 3 broad categories :

1) Static webpage
2) Dynamic webpage
3) Active webpage

Static documents :-

Static documents are fixed-content documents that a created and stored in a server. The client can get a copy of the document. In otherwords, the contents of the file are determined when the file is created, not when it is used. Static documents are prepared using one of the several languages : HTML, Extensible Markup language (XML), Extensible style language (XSL)

Dynamic documents :-

A dynamic document is created by webserver whenever browser requests a document. When a request arrives, the web server runs an application program or a script that creates a dynamic page. The server returns the result of the program or script as a response to the browser that requested the document. Because a fresh document is created for each request, the contents of the dynamic document may vary from one request to another.

A Very Simple example of a dynamic document
is the retrieval of time and date from Server.
The client gets a different dynamic document each time.

Active documents :-

For many applications, we need program or script to run at the client side. These are called active documents. For example Suppose we want to create animated graphics or the screen of program that interacts with the user. The program definitely needs to run at the client side where animations or interaction takes place. When a browser requests an active document, the server sends a copy of the document and a script.

The document is then run at the client side.
One way of creating an active document is to use java applets. It is compiled and ready to run The document is in byte code format. Another way is to use java scripts which are downloaded and run at client side

HTTP (hyper text transfer protocol):-

It is a protocol that is used to define how client server programs can be written to retrieve web pages from the the web. The server uses port number 80 at TCP, client uses a random port number. HTTP uses services of TCP. Before any transaction b/w client and server can take place, a connection needs to be established b/w them. After the transaction, the connection needs to be terminated. The server and dies, doesnot worry about errors in the messages exchanged or loss of any messages, because TCP is reliable and

Will take care of this matter.

Non persistent V/S persistent connections :-

If the web pages, objects to be retrieved are located on different servers, we dont have any other choice than to create a new TCP connection for retrieving each object. However if some of the objects are located in the same server we have two choices :

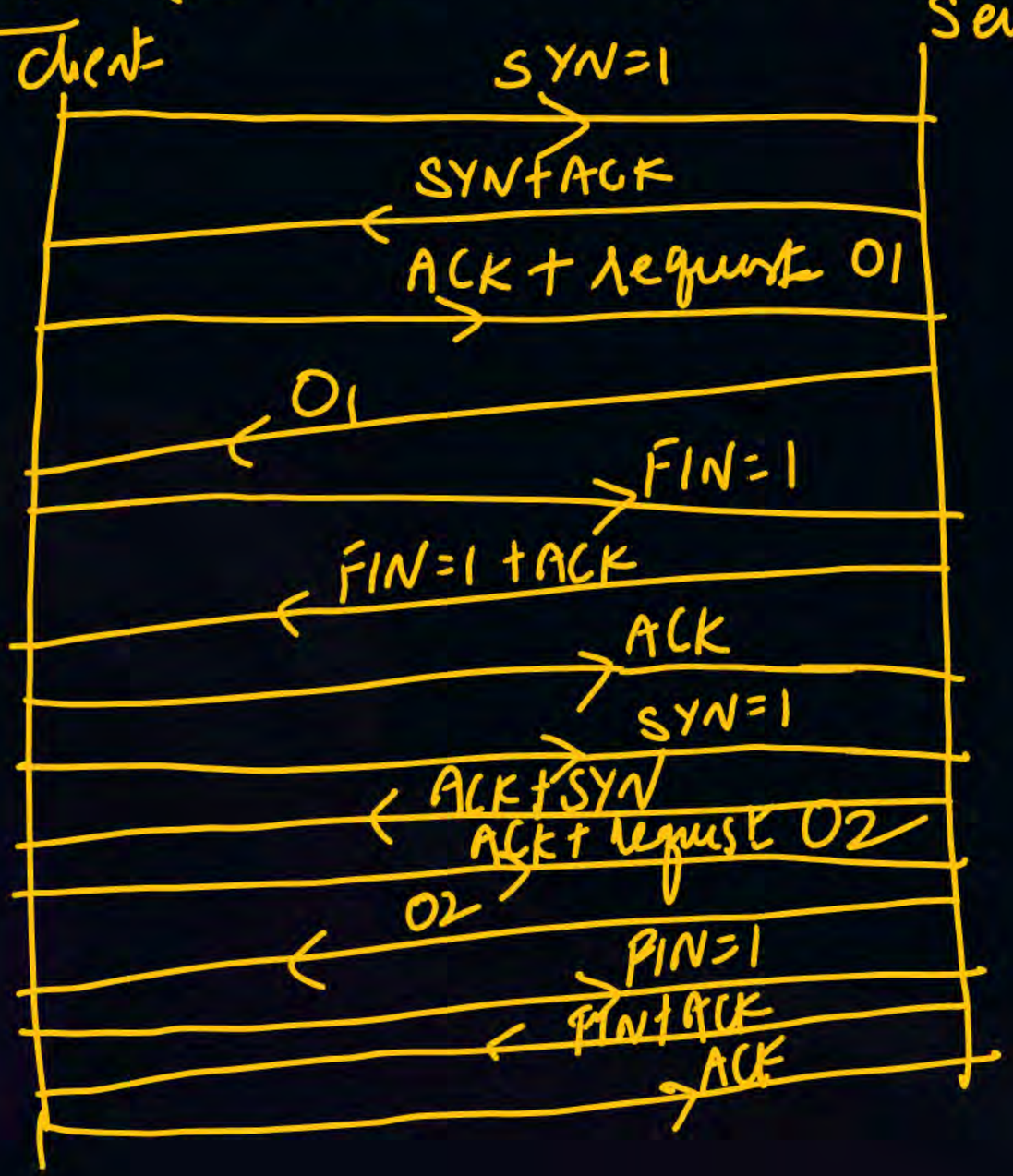
1) **Non persistent connections :-**

To retrieve each object, we use a new TCP connection

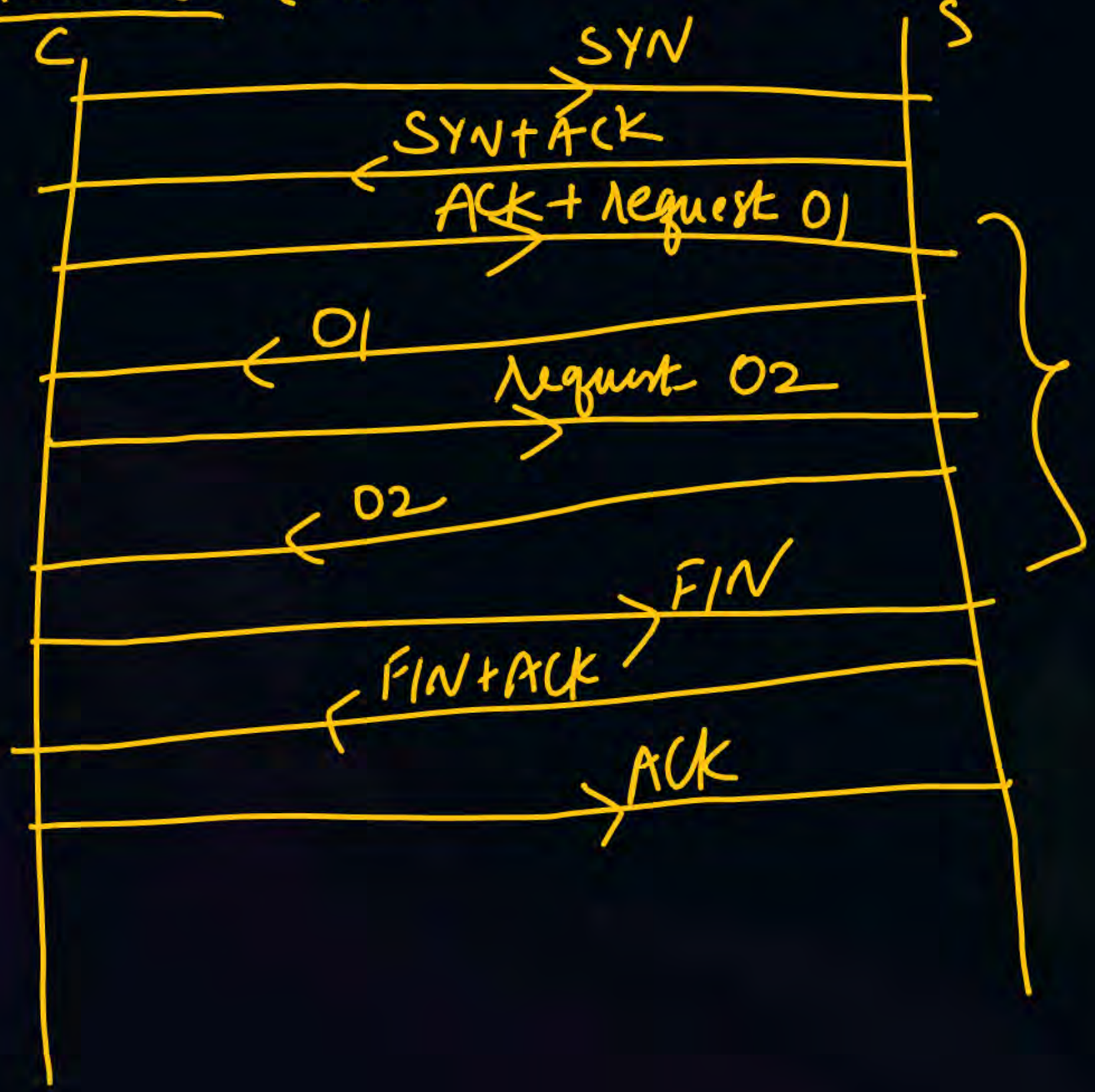
2) Persistent Connections :-

make a TCP connection and retrieve all the objects.

Non persistent :- (we need two objects from the server)



Persistant Connections :- (Assume we need 2 Objects



THANK - YOU